

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (IT)
December 2012

CE-712 Software Project Management & Quality Assurance (SPM & QA)

Date: 22.12.2012, Saturday Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Rough work is to be done in the last page of main supplementary, please don't write anything on the question paper.
5. Indicate clearly, the option(s) you attempt along with its respective question no.
6. Figures to the right indicate marks.

SECTION-I

Q-1 Answer the following questions.

1. What is a Project? What are the difference(s) between a Project and a Routine Work? What do you mean by Work Break-down Structure (WBS) in the context of a Software Projects? 3
2. Discuss the skills required by a Project Manager. What are the challenges and the problems faced by a Project Manager? 3
3. Explain various problems encountered with Software Projects. Explain the various activities covered by Software Project Management in detail. 3
4. It has been said that "The design cannot be directed by technical experts alone". Do you agree with this statement? Write in detail to support your view. 2

Q-2

- [A] Discuss about software quality factors and attributes. Discuss various software quality assurance activities. 4
- [B] You are a Project Manager, and you have met your full team for the first time. It is not what you expected. There are 15 inexperienced members and only 5 experienced members. You do not have the opinion of replacing any of them. Within the context of Project Management Strategy, what could you do? 4
- OR**
- [B] What can be done, if scope and requirements are changing continuously? This has severe implications on the projects. How can a Project Manager minimize their impacts on Projects? 4
- [C] List the common types of risks that a typical software project might suffer from. If you are the Project Manager of a large software development organization, what steps would you follow to manage risks in your software projects? 4

Q-3

- [A] Distinguish between software verification and software validation. When during the software life cycle are verification and validation performed? Can one be used in place of the other? 4
- [B] What are the various measures for failure of software projects? 4

OR

- [B] In the present scenario of frequent failure of software projects, can you put forward some strategies that facilitate successful software projects? 4
- [C] What are the principal activities involved in SCM (Software Configuration Management)? Why SCM is crucial to the success of a large software Project? 4

SECTION-II

Q-4

1. Explain the nature of various resources in detail. Categorize the required nature of resources and explain how is the cost monitored? 3
2. What is meant by Software Project Estimation? Explain different project estimation techniques. 4
3. Explain the components of a SPMP document. Explain the important issues that a project manager needs to document in a SPMP. 4

Q-5

- [A] Explain the components of a software Project progress. Describe about project progress and about the role of escalation. 4
- [B] Explain the steps involved in describing the problem. Explain the principles of software project decision process. 4
- [C] Explain the difference between Critical Path Analysis and Critical Chain Schedules? Describe CPM and PERT with suitable example. 4

OR

Q-5

- [A] Is critical path important if only one person is working on a software project? Discuss the concept of PERT/CPM in defining optimum schedule. 4
- [B] What are the five levels of CMM? Is it possible for an organization to achieve a higher level of CMM without achieving a lower one? 4
- [C] What is Project Quality Management? Discuss the output of the quality control process. 4

Q-6

- [A] Explain the software effort estimation techniques in detail. Justify that FPs give better estimate of software project size than LOC. 4

OR

- [A] Discuss about the initial Project Schedule Plan. Discuss the graphical notations which are used to illustrate the Project Schedule. 4
- [B] What are the needs of Project Management Methodologies? Indicate the levels of managerial efforts involved in project direction, planning, scheduling and controlling. 4
- [C] What is the role of computers in Project Management System? What are the things to be considered before using a software package for Project Management? 4

OR

- [C] Discuss the importance of a good time management for the successful completion of a software project. 4